

JAWAHARLAL NEHRU UNIVERSITY
SCHOOL OF COMPUTER AND SYSTEMS SCIENCES

Minor

M.C.A. (Monsoon Semester-2023)

Course Name: Computer Architecture

Maximum Marks: 30

Course Code: CS 405

Total Time: 1 hours

Q1. a) What is the difference between Von Neumann Architecture and Harvard Architecture?. What are the different types of addressing modes that has evolved in Computer Systems?.

b) A 32 bit floating point binary number has 4 bit plus sign for exponent and 26 plus sign for mantissa. The mantissa is a normalized function. Numbers in mantissa and exponent are in a signed-magnitude representation. What are the largest and smallest positive quantities that can be represented excluding zero?.

(10)

Q2. a) What is the difference between RISC and CISC type of Computer Architecture?. Explain with examples.

b) Which one is better and Why?. How can one compensate the drawbacks of one over the other type of architecture.

(10)

Q3. a) What is T- state, Machine Cycle and Instruction Cycle in a Computer System?.

b) Explain the concept of pipelining technique in a computer Architecture system?.

(10)